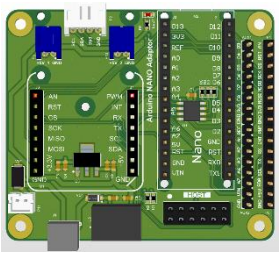
MAKE THINGS HAPPY !

Arduino Nano Adaptor

mikroBUS™ Adaptor for Arduino Nano Modules



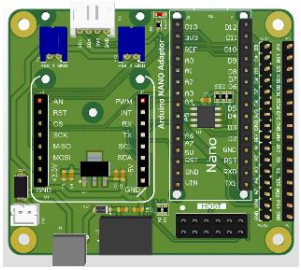
Arduino Nano Adaptor module (rev. 2.21) is designed for installing **Arduino Nano** format modules and **mikroBUS™** modules, as well as for connecting **IoTextra** series mezzanine modules via a cable.

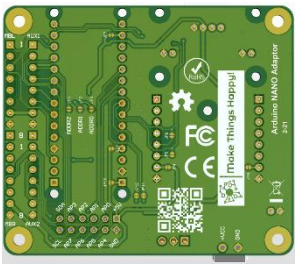
The module has an on-board **LM75B** temperature sensor connected to the **I²C** bus.

The module's input power is non-isolated, ranging from 9VDC to 36VDC. This power is supplied via a 2-pin terminal block with a 3.5mm pitch.

The module has the following slots:

- **NANO** (**M2** slot, **J8** and **J2** connectors) for installing **Arduino Nano** format modules
- **HOST** for connecting **IoTextra** series mezzanine modules via a cable
- **M1** slot (**J6** and **J7** connectors) compatible with **mikroBUS™**





The pinout of the **HOST** connector corresponds to the **HOST-P12** connector of the **IoTbase** series modules, so any **IoTextra** series mezzanine can be connected to the **Arduino Nano Adaptor** via a 12-pin cable.
The **I²C** interface is often used when designing devices and instruments based on the **Arduino Nano Adaptor**, so the module has an **I²C** connector (**P2**) for connecting external devices and sensors. By default, pull-up resistors for **I²C** are connected, but they can be disconnected.
The signals of the **NANO** slot that are not used by the **mikroBUS™** slot and the **HOST** connector are routed to the **AUX1** and **AUX2** auxiliary connectors.
The input voltage of 9VDC to 36VDC is converted to 5VDC by a converter. The maximum current is 1A. The module also has a linear regulator from 5VDC to 3.3VDC. The 5VDC voltage can be distributed to other modules via two connectors (**5V-1** and **5V-2**), and the input voltage via the **VIN** connector.
Since some of the modules installed in the **NANO** slot and in the **M1** slot can themselves generate 5VDC and 3.3VDC, the **Arduino Nano Adaptor** provides the ability to flexibly manage these connections using jumpers on its bottom side.
The module size is 65×56 mm. The module format corresponds to the popular **Raspberry Pi 3A+** board format, which greatly simplifies its use.
Primary applications of the module:

- Industry IoT
- Distributed Data Acquisition Systems
- Motor control
- Heating, Ventilation, & Air Condition Monitoring (HVAC)
- Smart sensors
- PLC
- Smart home
- Consumer electronics

QUICK START

The **Arduino Nano Adaptor** is most often used with one or another **IoTextra** series mezzanine, as together with a connected mezzanine and an **Arduino Nano** format module it is easy to use as a standalone device. The **Arduino Nano** format module is installed in the **NANO** slot, and the **IoTextra** mezzanine is connected to the **HOST** connector via a 12-pin cable (2×6, 2.54mm pitch, pin-to-pin connection 1–1 … 12–12).

Warning! You must pay attention to the orientation of the modules installed in the NANO slot. The USB connector of the installed module must face the edge of the board where the 5V-1, 5V-2 and I2C (P2) connectors are located. Incorrect installation can destroy the Arduino Nano Adaptor and Arduino compatible modules.

The following **IoTextra** series mezzanines can be connected to the **Arduino Nano Adaptor** via the **HOST** connector:

- **IoTextra Input**
- **IoTextra Relay2**
- **IoTextra SSR Small**
- **IoTextra MOSFET2**
- **IoTextra Octal, IoTextra Octal2, IoTextra Octal3 and IoTextra Octal4**
- **IoTextra Analog and IoTextra Analog3**
- **IoTextra Combo**

The current list of **IoTextra** series mezzanines is available on the website **http://www.makethingshappy.io**.

The **M1** slot allows the use of more than 2000 **MikroElektronika** **Click®** modules with the **mikroBUS™** bus:



To install **Click®** modules into the **mikroBUS™** slot, you don't need to do anything with the jumpers on the **Arduino Nano Adaptor** (you can use the default settings for all jumpers on the module).

Numerous sensors, peripherals, and modules can also be easily connected to the **Arduino Nano Adaptor** via the **I²C** connector (for example, **Grove** modules from Seeed Studio, using an appropriate cable):



DESCRIPTION

The **Arduino Nano Adaptor** has:

- **J8** and **J2** connectors of the **NANO** slot (**M2**)
- **J6** and **J7** connectors of the **M1** slot for the **mikroBUS™** bus
- **HOST** connector for connecting to **IoTextra** mezzanines
- **AUX1** and **AUX2** auxiliary connectors
- **MBL** (**P1**) and **MBR** (**P8**) connectors with the **mikroBUS™** bus signals (not installed, but can be self-installed by the user)
- **I2C** connector (**P2**) for the **I²C** bus
- **LM75B** temperature sensor
- **PWR** power connector (**J1**) for 9-36VDC power
- **VIN**, **5V-1** and **5V-2** connectors for power distribution

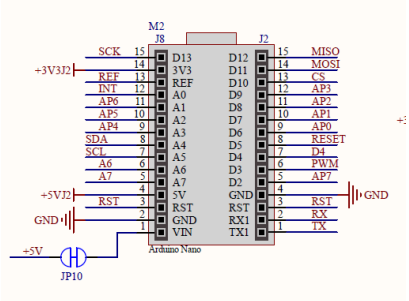
NANO slot. This slot consists of two 15-pin connectors (**J8** and **J2**). The following **Arduino Nano** compatible modules can be installed into this slot of the **Arduino Nano Adaptor**:

- **Arduino® Nano 33 IoT**
- **Arduino® Nano 33 BLE Rev2**

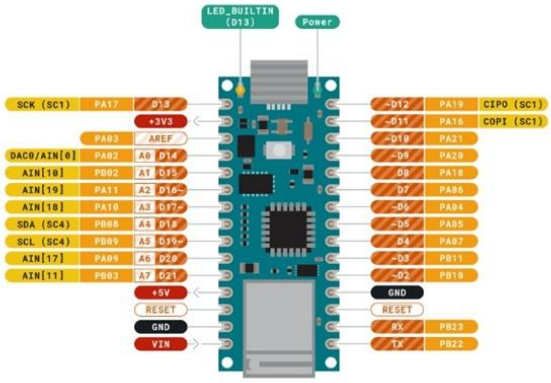
- **Arduino® Nano RP2040 Connect**
- **Arduino® Nano ESP32**
- **Arduino® Nano R4**
- **Arduino® Nano Matter**
- **Waveshare ESP32-S3-Nano**
- **STM32 Nucleo-32**

Attention! The module is incompatible with Arduino modules that have a 5V signal level (for example, the classic Arduino® Nano or Arduino® Nano Every).

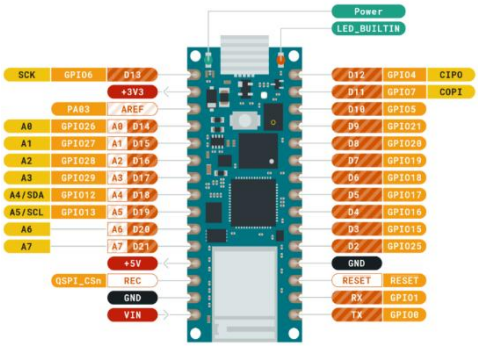
Slot structure:



As an example, we provide the pinout of the Arduino Nano 33 IoT and Arduino Nano RP2040 Connect:

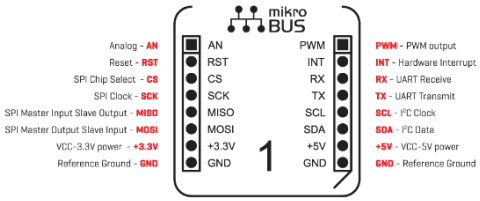


Arduino Nano 33 IoT pinout

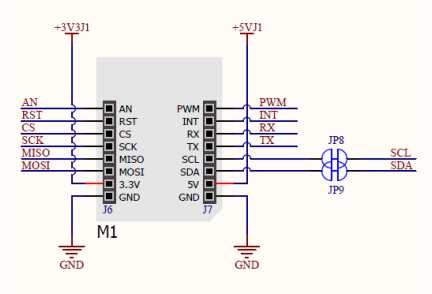


Arduino Nano RP2040 Connect pinout

mikroBUS™. The module has an **M1** slot (connectors **J6** and **J7**) which complies with the **mikroBUS™** standard:



The **M1** slot structure on the **Arduino Nano Adaptor**:



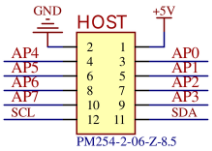
The **+3.3V** and **+5V** power of the **M1** slot can be disconnected from the module power using jumpers **JP6** and **JP4**, and the **SCL** and **SDA** signals can be disconnected from the common **I²C** bus using jumpers **JP8** and **JP9**.

Note. The AN signal of the M1 slot is not connected to the NANO slot (all NANO slot pins are used for other signals). The AN signal is available only on the MBL connector.

MBL and MBR Connectors. The **MBL** (**P1**) and **MBR** (**P8**) connectors (2.54mm, 1×8, not installed by default) duplicate all the **mikroBUS™** bus signals. Thus, through these connectors it is easy to connect another module (or even several) with **mikroBUS™** slots, increasing the number of slots on the common bus. Connector pinout:

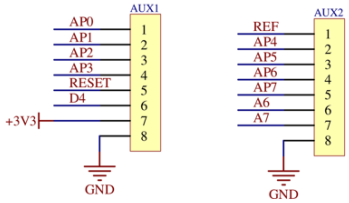
MBL pin	Signal	MBR pin	Signal
1	AN	1	PWM
2	RST	2	INT
3	CS	3	RX
4	SCK	4	TX
5	MISO	5	SCL
6	MOSI	6	SDA
7	+3V3	7	+5V
8	GND	8	GND

HOST Connector. This connector is for connecting application mezzanine modules (**IoTextra** series) to the **Arduino Nano Adaptor** via a 12-pin cable. The connector type is a 2×6 pin socket with a 2.54mm pitch. Connector pinout:



The **AP0**-**AP7** signals are general-purpose signals of the mezzanine. Their correspondence to the pins of the **NANO** slot is given in the **SIGNAL MATCHING** section.

AUX1 and AUX2 Auxiliary Connectors. Almost all signals of the **NANO** slot that are not used by the **mikroBUS™** slot are routed to the **AUX1** and **AUX2** connectors (2.54mm, 1×8), which makes it possible to use almost all **Arduino Nano** functions. The structure of these connectors is as follows:

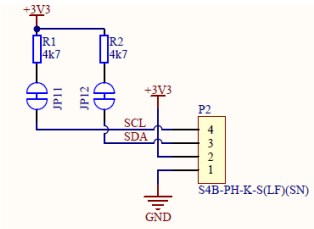


The **RESET** signal of the **AUX1** connector is connected to the **D5** pin of the **NANO** slot.

Attention! In rev. 2.21 the labels of the AUX1 and AUX2 connectors on the top side silkscreen do not correspond to the actual signals. Use the pinout given in this document:

AUX1 pin	Top silkscreen	Actual signal	AUX2 pin	Top silkscreen	Actual signal
1	D3	AP0 (D6)	1	REF	REF
2	D4	AP1 (D7)	2	A1	AP4 (A3)
3	D5	AP2 (D8)	3	A2	AP5 (A2)
4	D6	AP3 (D9)	4	A3	AP6 (A1)
5	D7	RESET (D5)	5	A6	AP7 (D2)
6	D8	D4	6	A7	A6
7	3V3	+3V3	7	VIN	A7
8	GND	GND	8	GND	GND

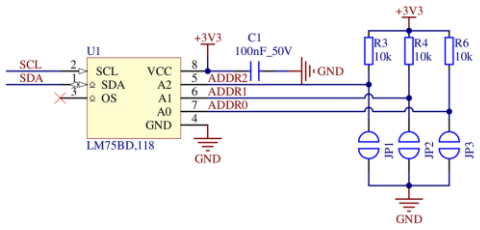
I²C Connector. The **I²C** connector (**P2**) is a 4-pin JST PH connector with a 2.0mm pitch (**S4B-PH-K-S**) and has the following structure:



By default, 4.7 kOhm pull-up resistors are connected to the **SCL** and **SDA** signals (jumpers **JP11** and **JP12**).

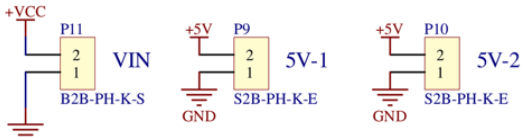
Temperature Sensor. The module has a temperature sensor - an **LM75B** chip, connected via **I²C**. Temperature measurement accuracy: ±2°C for the temperature range from −25°C to +100°C, ±3°C for the temperature range

from −55°C to +125°C. The **LM75B** chip can be replaced by the user with the **TMP1075** chip (maximum accuracy ±1°C).
The **I²C** address of the **LM75B** is 1001111x (0x4F) by default. This address can be changed by the user using jumpers **JP1**, **JP2** and **JP3** on the bottom side. The **OS** output of the sensor is not used.



POWER SUPPLY

The module's input power is non-isolated, ranging from 9VDC to 36VDC. This power is supplied via a 2-pin terminal block with a 3.5mm pitch (**PWR** connector, labeled **J1** on the module's schematic). Protection against incorrect input power polarity (Schottky diode **DSK34**) and a transient voltage suppressor (**SMAJ40CA**) are provided.
The input voltage of 9VDC to 36VDC is converted to 5VDC by a **K7805M-1000R3** converter. The maximum current is 1A.
The module also has a linear regulator from 5VDC to 3.3VDC (**AMS1117-3.3**). The total current consumed from the 5VDC (including the 3.3VDC consumers) must not exceed 1A.
The presence of 5VDC is indicated by a green **LED1**.
The 5VDC voltage can be distributed to other modules via the **5V-1** (**P9**) and **5V-2** (**P10**) connectors, and the input voltage (before the polarity protection diode) via the **VIN** (**P11**) connector. All these connectors are 2-pin JST PH connectors with a 2.0mm pitch:



By default, the 5VDC of the module is connected to the **5V** and **VIN** pins of the **NANO** slot (jumpers **JP5** and **JP10**), and the 3.3VDC is connected to the **3V3** pin of the **NANO** slot (jumper **JP7**). If the installed **Arduino Nano** format module generates its own 3.3VDC, it is recommended to open jumper **JP7**.

SIGNAL MATCHING

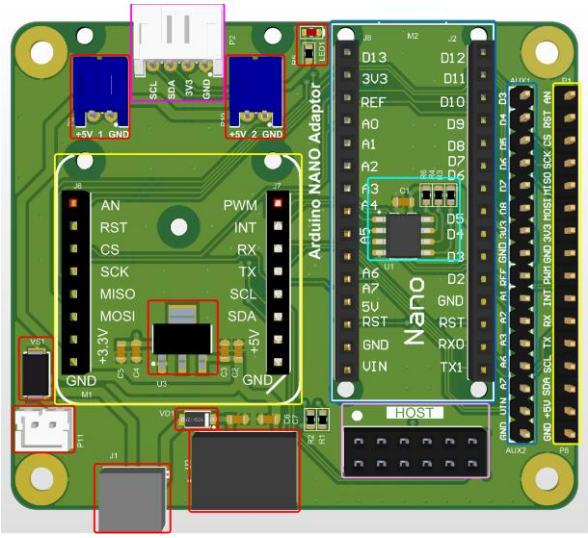
The **Arduino Nano Adaptor** combines the **Arduino Nano**, **mikroBUS™** and **HOST** (for communication with mezzanines) buses. The correspondence between the signals of the various buses in the **Arduino Nano Adaptor** is shown in the following table:

	Schema	NANO			mikroBUS (M1)		MBL / MBR	HOST	AUX1	AUX2
		J8	J2	Nano	LEFT	RIGHT				
UART	RX		2	RX0 / D0		RX	MBR-3			
	TX		1	TX1 / D1		TX	MBR-4			
I2C	SDA	8		A4		SDA*	MBR-6	11		
	SCL	7		A5		SCL*	MBR-5	12		
SPI	SCK	15		D13	SCK		MBL-4			
	MISO		15	D12	MISO		MBL-5			
	MOSI		14	D11	MOSI		MBL-6			
	CS		13	D10	CS		MBL-3			
RST	RST	3	3	RST	RST		MBL-2			
	RESET		8	D5					5	
PWM	PWM		6	D3		PWM	MBR-1			
INT & AN	INT	12		A0		INT	MBR-2			
	AN			—	AN		MBL-1			
	REF	13		AREF						1
AP0-AP7	AP0		9	D6				3	1	
	AP1		10	D7				5	2	
	AP2		11	D8				7	3	
	AP3		12	D9				9	4	
	AP4	9		A3				4		2
	AP5	10		A2				6		3
	AP6	11		A1				8		4
	AP7		5	D2				10		5
OTHER	D4		7	D4					6	
	A6	6		A6						6
	A7	5		A7						7

* The **SCL** and **SDA** signals of the **M1** slot are connected to the common **I²C** bus via jumpers **JP8** and **JP9** (closed by default).
The **RX** and **TX** signals of the **NANO** slot are connected directly to the **RX** and **TX** pins of the **mikroBUS™** slot.
The **AN** signal of the **mikroBUS™** slot is connected only to the **MBL** connector.

LAYOUT

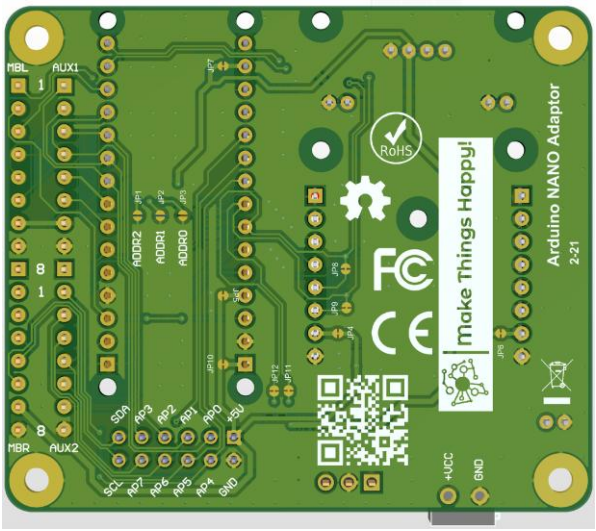
Below is the layout of the elements on the top side of the **Arduino Nano Adaptor**:



In this picture:

- **Red** highlights power-related elements (external PWR power connector J1, VIN, 5V-1 and 5V-2 connectors, VCC -> 5VDC converter, 5VDC -> 3.3VDC regulator, protection elements, and LED for 5VDC power indication)
- **Blue** highlights the NANO slot connectors (J8 and J2) and the AUX1 and AUX2 auxiliary connectors
- **Yellow** highlights the mikroBUS™ slot M1 (J6 and J7) and the MBR and MBL connectors
- **Pink** highlights the HOST connector for communication with mezzanines
- **Purple** highlights the I2C connector for connecting peripherals via the I²C bus
- **Light blue** highlights the LM75B temperature sensor

The layout of the elements on the bottom side of the **Arduino Nano Adaptor** is provided next.



JUMPERS

All **Arduino Nano Adaptor** jumpers are located on the bottom side:

1) **JP1, JP2 and JP3**. For selecting the address of the **LM75B** temperature sensor on the **I²C** bus. By default, the address is 1001111x (all jumpers are open). The table for selecting the address is provided below.

JP1 (A2)	JP2 (A1)	JP3 (A0)	LM75B address
CLOSE	CLOSE	CLOSE	1001000x
CLOSE	CLOSE	OPEN	1001001x
CLOSE	OPEN	CLOSE	1001010x
CLOSE	OPEN	OPEN	1001011x
OPEN	CLOSE	CLOSE	1001100x
OPEN	CLOSE	OPEN	1001101x
OPEN	OPEN	CLOSE	1001110x
OPEN	OPEN	OPEN	1001111x

2) **JP4 and JP6**. They are designed to disconnect the +5V power (jumper **JP4**) and +3V3 power (jumper **JP6**) from the **M1** slot with the **mikroBUS™** bus on the module. By default, power is connected (CLOSE).
3) **JP5 and JP7**. They are designed to disconnect the +5V power (jumper **JP5**) and +3V3 power (jumper **JP7**) from the **5V** and **3V3** pins of the **NANO** slot. By default, power is connected (CLOSE).
4) **JP8 and JP9**. They are designed to disconnect the **SCL** (jumper **JP8**) and **SDA** (jumper **JP9**) signals of the **M1** slot from the common **I²C** bus. By default, the signals are connected (CLOSE).
5) **JP10**. Determines whether the module's +5V is connected to the **VIN** pin of the **NANO** slot. By default, it is CLOSE, meaning it is connected.
6) **JP11 and JP12**. For connecting 4.7 kOhm pull-up resistors to the **SCL** (jumper **JP11**) and **SDA** (jumper **JP12**) signals of the **I²C** bus of the **Arduino Nano Adaptor**. By default, they are CLOSE, meaning the resistors are connected.

Summary of the default jumper states:

Jumper	Function	Default
JP1-JP3	LM75B I²C address (A2, A1, A0)	OPEN
JP4	+5V to M1 (mikroBUS™)	CLOSE
JP5	+5V to 5V pin of NANO slot	CLOSE
JP6	+3V3 to M1 (mikroBUS™)	CLOSE
JP7	+3V3 to 3V3 pin of NANO slot	CLOSE
JP8	SCL of M1 to common I²C bus	CLOSE
JP9	SDA of M1 to common I²C bus	CLOSE
JP10	+5V to VIN pin of NANO slot	CLOSE
JP11	SCL pull-up 4.7 kOhm	CLOSE
JP12	SDA pull-up 4.7 kOhm	CLOSE

MEZZANINES

Any mezzanine from the **IoTextra** series listed on the website **http://www.makethingshappy.io** can be used as a mezzanine. The mezzanine is connected to the **HOST** connector of the **Arduino Nano Adaptor** via a 12-pin cable (see **QUICK START**).

ACCESSORIES

The following accessories may be required to use the module:

- Mating (plug) two-pin connector with a 3.5mm pitch for module power
- Two-pin cables with JST PH connectors for the **VIN**, **5V-1** and **5V-2** connectors
- **I²C** cable with a four-pin JST PH connector
- 12-pin cable (2×6, 2.54mm pitch) for connecting the **Arduino Nano Adaptor** to **IoTextra** series mezzanines
- Two 8-pin headers with a 2.54mm pitch for the **MBL** and **MBR** connectors